

Delhivery - Business Analysis Report

 Business_Case_Delhivery_Feature_Engineering.pdf

<https://drive.google.com/file/d/1y4NLqQXb-hkwbimjGlmRqqJDshOlrvCa/view?usp=sharing>

Business Recommendations (actionable insights)

1. **Focus on top corridors:** Given the concentration of volume, prioritize resources (trucks, scheduling) on the busiest corridors. For example, the Bangalore inner-city routes (Nelmngla↔Bomsndra↔Airport) and the Gurgaon-Haryana hub are critical – scheduling more frequent shuttles or allocating larger vehicles there could improve efficiency.
2. **Improve ETA accuracy:** The actual vs OSRM gap (~2×) is large. Route planning and customer SLAs should incorporate this factor. Investigate why times are so inflated (e.g. traffic, loading delays, multi-stops) and adjust OSRM or routing parameters accordingly. For instance, add typical dwell times per stop or use historical data to better predict trip duration.
3. **Optimize multi-stop scheduling:** Since ~80% of trips are multi-segment, optimizing the sequence of stops (or considering dynamic re-routing) could cut down on cumulative delays. If certain intermediate centers cause heavy delays, consider direct runs or cross-docking alternatives.
4. **Address data issues:** Fix the anomalous data collection (negative and extremely high times) to avoid skewed analytics. This may involve calibrating scanners timestamps or removing obviously bad records from KPIs. Accurate timing is crucial for on-time performance metrics.
5. **State-level strategy:** Haryana (especially the Gurgaon hub) accounts for ~16% of trips. Investing in that region (e.g. additional hub capacity or local delivery fleet) could yield outsized benefits. Likewise, build capacity around Karnataka (Bengaluru) to handle its large share.

Actionable Business Items

1. Prioritize Key Routes

- Focus more resources (trucks, staff, planning) on **routes between Bengaluru and Gurgaon** — they carry the highest delivery volume.
- Run more **frequent shuttles** on these corridors to reduce delays and avoid congestion.

2. Plan Deliveries Around Delays

- Deliveries are taking **twice as long** as estimated — build this into your planning.
- Give customers **realistic delivery windows** that include buffer time for delays.

3. Fix Bad Data Entries

- Some trips have **impossible times** (like negative minutes or several days long).
- Set up a simple check to catch and remove these before they affect decisions.

4. Add More Trucks During Peak Hours

- Trips spike during certain hours (usually daytime).
- Use **extra vehicles and manpower** during busy periods to meet demand on time.

5. Streamline Multi-Stop Routes

- Many trips make multiple stops — and that's slowing things down.
- Group nearby stops or run **direct delivery routes** when possible to save time.

6. Give Special Attention to Top Hubs

- Hubs like **Gurgaon_Bilaspur** and **Bangalore_Nelmangla** handle the most trips.
- Improve infrastructure, staffing, or capacity at these locations to keep operations smooth.

7. Use Real Trip Time Instead of Google Maps

- Trips often take much longer than GPS estimates.
- Plan delivery schedules based on **actual past trips**, not just map estimates.

8. Watch Out for Long-Distance Trips

- Some deliveries go over 1,900 km — these need **special planning**.
- Assign them early, use FTL (Full Truck Load), and ensure clear tracking.

9. Train Teams on Delay Patterns

- Teach ground staff where delays often happen (e.g. specific cities or routes).
- Encourage proactive actions like calling ahead or loading faster.

10. Review State Performance

- Focus on **Haryana, Karnataka, and Maharashtra** — they make up most trips.
- Expand or improve logistics operations in these key states first.

Logistics Performance KPIs

KPI	Value	Interpretation
Total Trips	14,817 trips	Number of complete deliveries (multi-stop or single) handled
Total Segments	144,867 delivery legs	Each stop between source and destination counted individually
Avg Trip Distance	234 km	Typical end-to-end trip distance
Avg Trip Time	417 minutes (~7 hours)	Time taken from source to final delivery point
Avg Segment Time	36.2 minutes	Time taken for one delivery leg
Avg Delay Factor	2.12× longer than map estimate	Actual deliveries take more than double the estimated time (OSRM-based)
Top Route Type by Volume	FTL (Full Truck Load) – 68.8%	Most deliveries use full-truck assignments (vs. Carting)
Top Source Center	Gurgaon_Bilaspur_HB – 16.1%	Major origin point in Haryana
Top Destination Center	Bengaluru_KGAirprt_HB	Most popular delivery endpoint
Top Corridor	Bengaluru_Nelmngla → KGAirprt	Highest volume of delivery traffic
Fastest 25% of Trips (Q1 Time)	≤ 51 minutes	One-quarter of trips are very short and quick
Longest Trip Time Recorded	4,532 minutes (~75 hours)	Major outlier—likely needs investigation
Longest Distance Recorded	1,927 km	Outlier for long-distance trips
Share of Top 5 Source Centers	~37% of total trips	Indicates high concentration of activity at few hubs

